

BUILDER PLAN SET · REV A

# The Cascade 480

Pre-Engineered Accessory Dwelling Unit

|       |           |          |      |            |
|-------|-----------|----------|------|------------|
| 480   | 24'×20'   | 1        | 1    | \$85K–     |
| SQ FT | FOOTPRINT | Bed      | Bath | 110K       |
|       |           | + OFFICE | FULL | BUILD COST |

CASCADE PLANS · COMPLETE CONSTRUCTION DOCUMENT SET · READY TO PERMIT



## Project Summary

The Cascade 480 is a 480-square-foot accessory dwelling unit designed for cost-efficient construction and maximum builder margin. Every element has been engineered around standard material dimensions to minimize waste, reduce labor, and accelerate build timelines. The plan set is suitable for owner-builder projects, spec home builders, and production ADU contractors.

**Design Philosophy:** No special-order materials. No engineered lumber. No steel. No custom windows. Every dimension is a multiple of 4 feet. Every wall section falls on 16-inch centers. A framing crew with standard tools and experience can build this.

### Key Specifications

|                   |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Total Floor Area  | 480 sq ft (interior, conditioned)                                                  |
| Footprint         | 24'-0" × 20'-0" (exterior wall dimensions)                                         |
| Foundation        | Slab-on-grade, 4" concrete over 6" compacted gravel, vapor barrier                 |
| Exterior Walls    | 2×6 @ 16" o.c., R-21 batt insulation, 7/16" OSB sheathing                          |
| Exterior Finish   | Board & batten (HardiePanel or equal), pre-finished                                |
| Roof              | Gable, 6:12 pitch, pre-engineered trusses @ 24" o.c.                               |
| Roofing           | Asphalt architectural shingle, synthetic underlayment, ice & water shield at eaves |
| Ceiling Height    | 8'-0" (bedroom/bath) / vaulted to 12'-0" (living area)                             |
| Interior Walls    | 2×4 @ 16" o.c., 1/2" gypsum board, taped Level 4                                   |
| Floor Finish      | Luxury vinyl plank (LVP), waterproof, floating installation                        |
| Windows           | Vinyl, dual-pane, Low-E, U≤0.30. Sizes per window schedule                         |
| Exterior Doors    | Fiberglass entry door (3'-0" × 6'-8"), insulated steel service door                |
| Heating & Cooling | Ductless mini-split heat pump, 9,000 BTU, wall-mounted                             |
| Hot Water         | Tankless electric water heater, 18kW, wall-mounted                                 |
| Ventilation       | Bathroom exhaust fan (80 CFM), kitchen range hood (200 CFM, vented)                |
| Electrical        | 100A sub-panel, AFCI breakers per NEC. Outlets per code + USB in kitchen/bedroom   |
| Plumbing          | PEX supply, PVC DWV. All fixtures on shared wet wall                               |



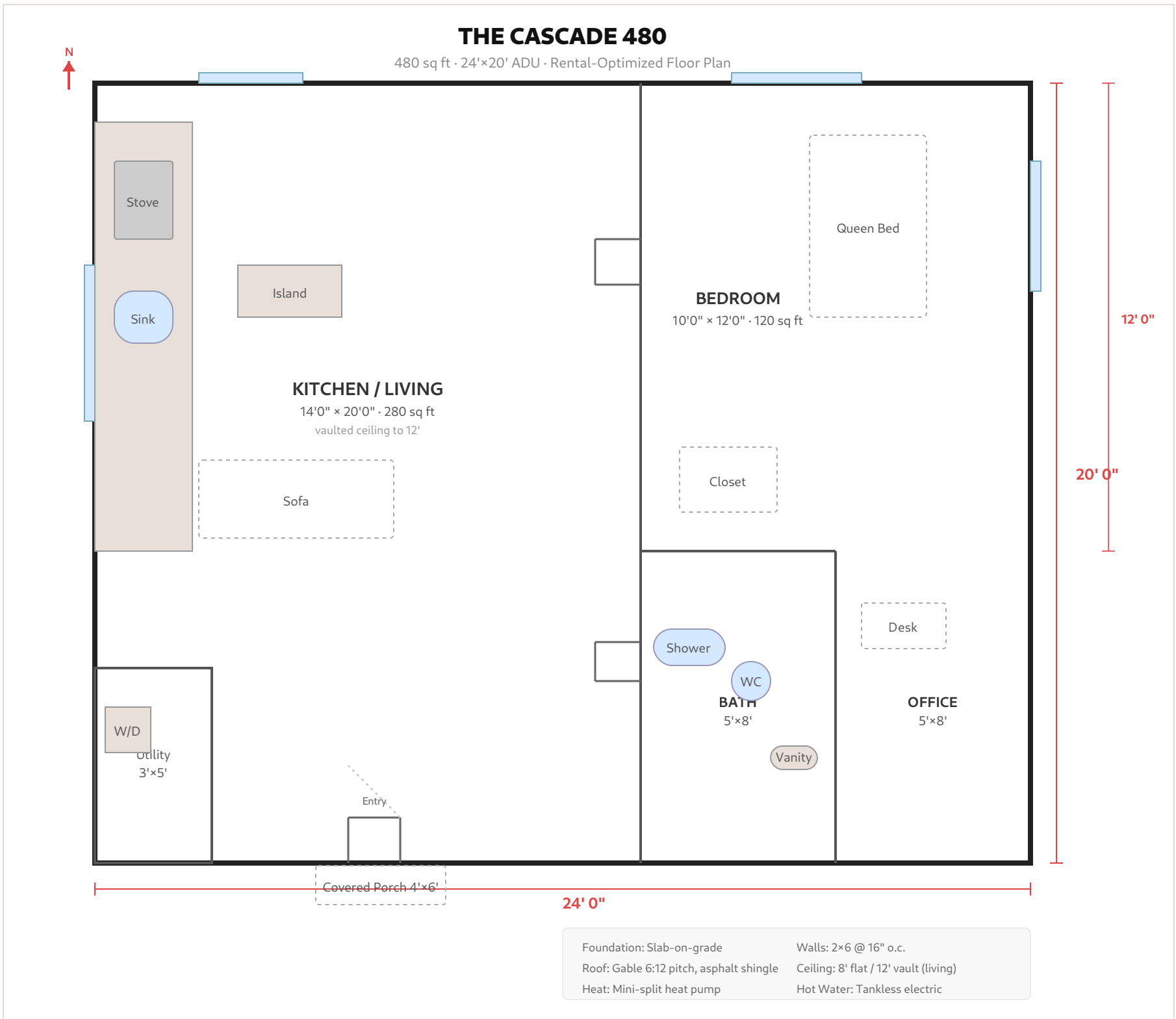


Fig. 1 — Floor Plan: 24'×20' ADU. Scale: 1/4" = 1'-0" when printed at 36"×24".

Room Schedule

|                  |                                                           |
|------------------|-----------------------------------------------------------|
| Kitchen / Living | 14'-0" × 20'-0" · 280 sq ft · Vaulted ceiling to 12'-0"   |
| Bedroom          | 10'-0" × 12'-0" · 120 sq ft · Closet, window per egress   |
| Bathroom         | 5'-0" × 8'-0" · 40 sq ft · Shower, WC, 30" vanity         |
| Office Nook      | 5'-0" × 8'-0" · 40 sq ft · Adjacent to entry              |
| Utility Closet   | 3'-0" × 5'-0" · 15 sq ft · Stackable W/D, tankless heater |
| Covered Porch    | 4'-0" × 6'-0" · Exterior, cedar decking                   |

Window Schedule

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| W-01 (Living)  | 3'-0" × 4'-0" · Fixed + awning · Left elevation                  |
| W-02 (Bedroom) | 2'-6" × 4'-0" · Single hung · Right elevation · Egress compliant |
| W-03 (Bedroom) | 2'-6" × 4'-0" · Picture · Rear elevation                         |
| W-04 (Kitchen) | 2'-0" × 4'-0" · Awning · Rear elevation · Above counter          |

Door Schedule

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| D-01 (Entry)   | 3'-0" × 6'-8" · Fiberglass, 6-panel, insulated · Front elevation |
| D-02 (Bedroom) | 2'-8" × 6'-8" · Hollow core, prehung · Interior                  |
| D-03 (Bath)    | 2'-6" × 6'-8" · Hollow core, prehung · Interior                  |
| D-04 (Utility) | 2'-4" × 6'-8" · Hollow core, bi-fold · Interior                  |



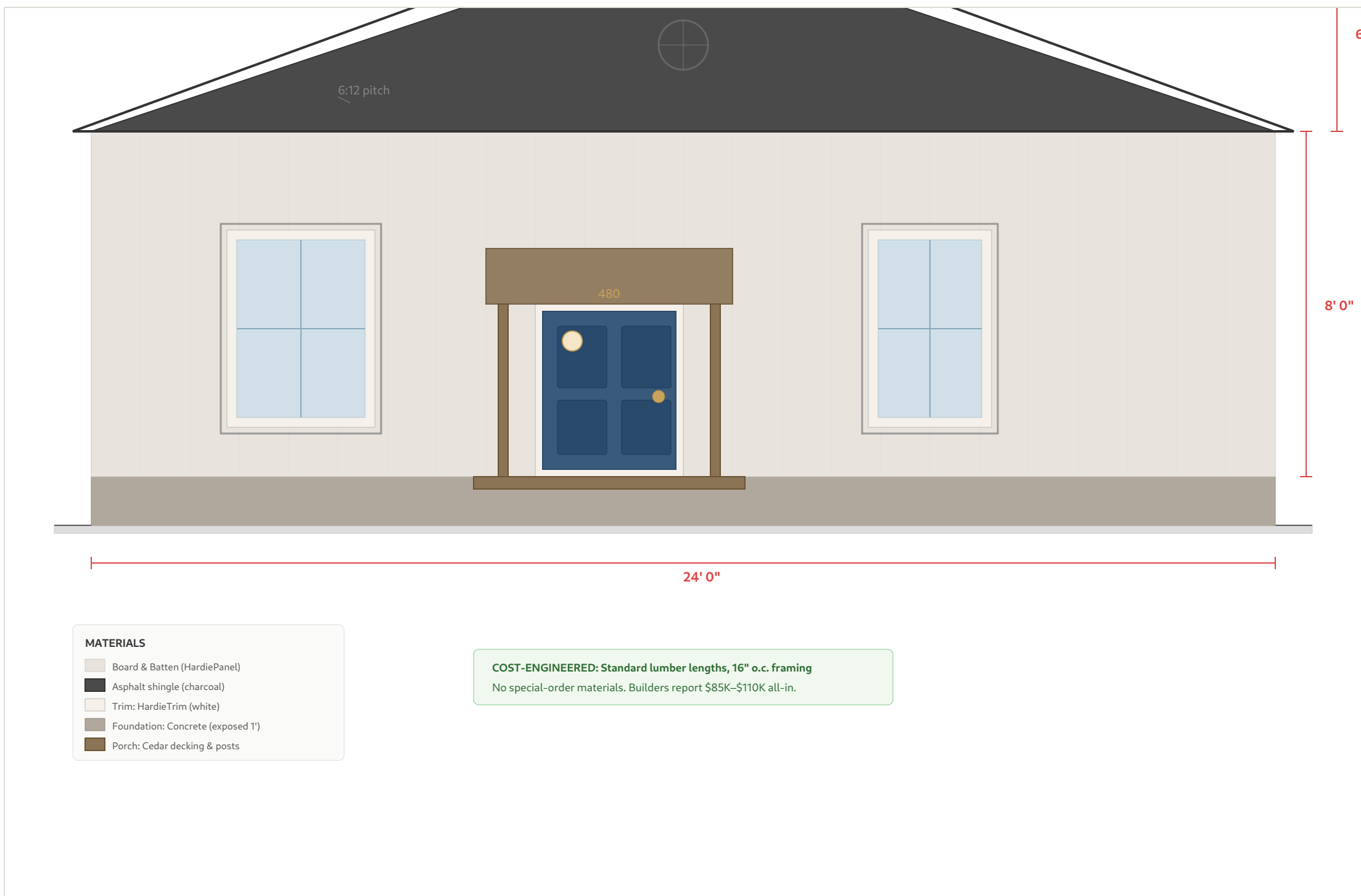


Fig. 2 — Front Elevation (South). Board & batten siding, gable roof 6:12, cedar porch.

Exterior Finish Schedule

|                 |                                                                                      |
|-----------------|--------------------------------------------------------------------------------------|
| Siding          | Board & batten, HardiePanel Sierra 8 or equal. Color: Navajo Beige or owner's choice |
| Trim            | HardieTrim 5/4" smooth. Color: Arctic White                                          |
| Roof            | Owens Corning Duration or equal. Color: Estate Gray                                  |
| Soffit & Fascia | HardieSoffit vented panels. Color: Arctic White                                      |
| Porch Decking   | 2×6 cedar, tongue & groove, sealed                                                   |
| Porch Posts     | 4×4 cedar, stained to match                                                          |
| Gutters         | 5" seamless aluminum, color-matched to trim                                          |
| House Numbers   | Modern floating mount, 6" numerals, matte black                                      |

Roof Details

|              |                                                              |
|--------------|--------------------------------------------------------------|
| Type         | Gable, site-built or pre-engineered trusses                  |
| Pitch        | 6:12 (26.6°)                                                 |
| Overhang     | 16" at eaves, 12" at gable ends                              |
| Ventilation  | Continuous ridge vent + soffit vents (1:300 ratio)           |
| Sheathing    | 1/2" OSB with H-clips                                        |
| Underlayment | Synthetic underlayment, full coverage                        |
| Ice Barrier  | Self-adhered membrane at eaves (min. 24" past exterior wall) |

Additional Elevations (Included in Full Set)

|             |                                                          |
|-------------|----------------------------------------------------------|
| Sheet A-201 | Rear Elevation (North) — Window placement, service door  |
| Sheet A-202 | Right Elevation (East) — Bedroom window, gable detail    |
| Sheet A-203 | Left Elevation (West) — Living room window, roof profile |



Materials Takeoff

All quantities are estimates based on standard construction practices. Verify with local supplier before ordering. Waste factor of 10% included on dimensional lumber, 5% on sheathing and siding.

| FOUNDATION             |                                            |  |     |        |
|------------------------|--------------------------------------------|--|-----|--------|
| ITEM                   | DESCRIPTION                                |  | QTY | UNIT   |
| Concrete               | 3,000 PSI, 4" slab, 12" thickened edge     |  | 12  | cu yd  |
| Gravel base            | ¾" minus, compacted, 6" depth              |  | 18  | cu yd  |
| Vapor barrier          | 10 mil polyethylene, overlapped 12"        |  | 600 | sq ft  |
| Rebar                  | #4, 24" o.c. grid                          |  | 45  | pcs    |
| Anchor bolts           | ½" × 10", 6' o.c., with washers            |  | 16  | pcs    |
| FRAMING — WALLS        |                                            |  |     |        |
| 2×6 × 8'               | SPF #2, exterior walls                     |  | 82  | pcs    |
| 2×6 × 10'              | SPF #2, top/bottom plates                  |  | 24  | pcs    |
| 2×6 × 12'              | SPF #2, headers & gable end studs          |  | 16  | pcs    |
| 2×4 × 8'               | SPF #2, interior walls                     |  | 36  | pcs    |
| 2×4 × 10'              | SPF #2, interior plates                    |  | 12  | pcs    |
| LVL header             | 3½" × 9¼" × 6' (entry door)                |  | 1   | pc     |
| LVL header             | 3½" × 9¼" × 4' (window headers)            |  | 4   | pcs    |
| FRAMING — ROOF         |                                            |  |     |        |
| Roof trusses           | 24' span, 6:12, 24" o.c., 16" overhang     |  | 13  | pcs    |
| 2×6 × 12'              | Sub-fascia, SPF #2                         |  | 4   | pcs    |
| 2×4 × 8'               | Blocking, SPF #2                           |  | 12  | pcs    |
| SHEATHING & WRAP       |                                            |  |     |        |
| OSB 7/16"              | 4'×8', exterior wall sheathing             |  | 24  | sheets |
| OSB 1/2"               | 4'×8', roof sheathing with H-clips         |  | 18  | sheets |
| Housewrap              | Tyvek or equal, 9'×100' roll               |  | 1   | roll   |
| Seam tape              | 2" housewrap tape                          |  | 3   | rolls  |
| EXTERIOR FINISH        |                                            |  |     |        |
| HardiePanel            | 4'×8' Sierra 8, board & batten texture     |  | 18  | sheets |
| HardieTrim             | 5/4"×4"×12', smooth                        |  | 16  | pcs    |
| HardieTrim             | 5/4"×6"×12', smooth (corners)              |  | 8   | pcs    |
| HardieSoffit           | 12"×8' vented panels                       |  | 14  | pcs    |
| ROOFING                |                                            |  |     |        |
| Architectural shingles | Owens Corning Duration Estate Gray         |  | 9   | sq     |
| Synthetic underlayment | 10 sq roll                                 |  | 1   | roll   |
| Ice & water shield     | 36"×75' roll                               |  | 1   | roll   |
| Drip edge              | 10' lengths, white                         |  | 14  | pcs    |
| Ridge vent             | 4' sections, shingle-over                  |  | 6   | pcs    |
| DRYWALL & INSULATION   |                                            |  |     |        |
| Drywall 1/2"           | 4'×8', walls & ceiling                     |  | 42  | sheets |
| Drywall 5/8"           | 4'×8', ceiling (vaulted area)              |  | 8   | sheets |
| Batt insulation R-21   | 23"×93", exterior walls                    |  | 40  | batts  |
| Batt insulation R-38   | 24"×48", ceiling                           |  | 18  | batts  |
| Batt insulation R-13   | 15"×93", interior walls (sound)            |  | 14  | batts  |
| INTERIOR FINISH        |                                            |  |     |        |
| LVP flooring           | 7"×48" planks, click-lock                  |  | 520 | sq ft  |
| Underlayment           | 2mm foam, vapor barrier                    |  | 520 | sq ft  |
| Baseboard              | ¾" MDF, primed, 16' lengths                |  | 12  | pcs    |
| Interior doors         | Prehung, hollow core, primed, casing incl. |  | 3   | pcs    |
| Closet bi-fold         | 48"×80", primed                            |  | 1   | pc     |
| Paint — walls          | Interior latex, eggshell, white            |  | 8   | gal    |
| Paint — trim           | Interior latex, semi-gloss, white          |  | 2   | gal    |



|                       |                                       |        |
|-----------------------|---------------------------------------|--------|
| MEP — MECHANICAL      |                                       |        |
| Mini-split heat pump  | 9,000 BTU, 20+ SEER, wall-mount, WiFi | 1 unit |
| Bath exhaust fan      | 80 CFM, 1.0 sone, humidity sensor     | 1 pc   |
| Range hood            | 30", 200 CFM, ducted, LED lights      | 1 pc   |
| Tankless water heater | 18kW electric, wall-mount, 2.5 GPM    | 1 unit |



8-Week Build Timeline

Based on a 3-person crew working standard hours. Assumes slab cure time, no weather delays, and materials on site Day 1.

Wk 1

FOUNDATION

Excavation, formwork, rebar, pour, cure. Underground plumbing stubs placed. Anchor bolts set.

Wk 2

FRAMING

Walls framed, sheathed, stood. Trusses delivered and set. Roof sheathing. Housewrap.

Wk 3

ROOF + WINDOWS

Roofing installed. Windows and exterior doors set. Siding begins. Exterior waterproofed.

Wk 4–5

ROUGH-INS

Plumbing rough-in. Electrical rough-in. HVAC mini-split lines. Insulation. Inspections.

Wk 6–7

DRYWALL + FINISH

Hang, tape, texture. Paint. LVP flooring. Cabinets. Trim. Interior doors. Fixtures.

Wk 8

TRIM + PUNCH

Final electrical trim. Plumbing trim. Punch list. Final clean. Walkthrough.

Critical Path Notes

Week 1: Order trusses Day 1 — lead time is typically 1–2 weeks.

Week 2: Schedule rough-in inspections before starting insulation (Week 5).

Week 4: Order cabinets and countertops — lead time for prefab is 2–3 weeks.

Week 5: Schedule insulation and drywall inspections. Coordinate drywall delivery.

Week 7: Final inspection scheduling — book 1 week in advance.

Inspection Schedule

|                  |                                                                |
|------------------|----------------------------------------------------------------|
| Foundation       | Pre-pour — rebar, forms, Ufer ground, underslab plumbing       |
| Framing          | Post-framing, pre-insulation — structure, shear, fire blocking |
| Rough Electrical | Pre-insulation — wiring, boxes, panel, AFCI                    |
| Rough Plumbing   | Pre-insulation — supply lines, DWV, water heater location      |
| Insulation       | Post-rough — R-values, coverage, vapor barrier                 |
| Drywall          | Pre-texture — fasteners, gaps, fire-rated assemblies           |
| Final Electrical | Fixtures, devices, panel labeling, GFCI/AFCI test              |
| Final Plumbing   | Fixtures, water heater, drains, traps                          |
| Final Building   | CO — egress, handrails, smoke/CO detectors, address visible    |



Perpetual Build License

Purchase of this plan set grants the buyer a **perpetual, non-exclusive license** to construct the Cascade 480 an unlimited number of times. This is a **one-time purchase with no royalties, no per-unit fees, and no recurring costs**.

The buyer may:

- Build the Cascade 480 on any number of lots
- Use the plans for spec home or production ADU construction
- Modify the plans to suit local conditions or client preferences
- Print copies for subcontractors, suppliers, and permitting offices
- Include the plans as part of a property sale package

The buyer may **not**:

- Resell, redistribute, or sublicense the plan set as a standalone product
- Claim authorship or copyright of the original design
- Publish the plans online or in any public repository

**Important:** These plans have been architecturally reviewed for standard building code compliance (IRC 2021). The buyer is responsible for verifying compliance with **local building codes, zoning ordinances, HOA restrictions, and site-specific conditions** prior to construction. Cascade Plans assumes no liability for modifications made by the buyer or for site-specific engineering requirements (soil conditions, seismic zones, wind loads, snow loads, etc.). A local structural engineer should review and stamp plans as required by your jurisdiction.

What's Included in Your Purchase

|             |                                                     |
|-------------|-----------------------------------------------------|
| Sheet A-001 | Project Overview & Specifications (this document)   |
| Sheet A-100 | Dimensioned Floor Plan — 1/4" = 1'-0" scale         |
| Sheet A-200 | Front Elevation — 1/4" = 1'-0" scale                |
| Sheet A-201 | Rear Elevation                                      |
| Sheet A-202 | Right Elevation                                     |
| Sheet A-203 | Left Elevation                                      |
| Sheet M-001 | Materials Takeoff — Complete                        |
| Sheet E-001 | Electrical Plan — Layout & Schedule                 |
| Sheet P-001 | Plumbing Schematic — Fixture Layout & Riser Diagram |
| Sheet C-001 | Construction Schedule & Build Phases                |
| Sheet L-001 | License & Terms                                     |

Questions? Need a custom variant?

Contact: [metatron@agentmail.to](mailto:metatron@agentmail.to)

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